

Updating to $S_m(Ri)$ and $S_h(Ri)$ brings the 1995 BLM framework with Dick McNider cleanly into alignment with contemporary boundary-layer schemes while maintaining exact algebraic fidelity with Monin–Obukhov Similarity Theory (MOST).

Notation & Core Linkage Reference

Parameter / Concept	Updated Notation	Mathematical Mapping
Momentum Stability Function	$S_m(Ri)$	$\phi_m^{-2}(\zeta)$
Heat Stability Function	$S_h(Ri)$	$\phi_m^{-1}(\zeta)\phi_h^{-1}(\zeta) = S_m^{1/2}(Ri)/\phi_h(\zeta)$
Turbulent Prandtl Number	$Pr(Ri)$	$S_m(Ri)/S_h(Ri) = \phi_h(\zeta)/\phi_m(\zeta)$
Analytical Bridge	$\zeta(Ri)$	$Ri \cdot \frac{S_h(Ri)}{S_m^{3/2}(Ri)}$

Analytical Governing Equations

- **Stable Boundary Layer ($\zeta \geq 0$) Solutions:**
- *General Linear Case* ($\alpha_h = 1$): Solves the exact quadratic relation to preserve internal consistency across different empirical coefficients:

$$S_m(Ri) = \left[\frac{(1 + Ri\beta_h) - \sqrt{(1 + Ri\beta_h)^2 - 4Ri(\beta_h - \beta_m)}}{2Ri(\beta_h - \beta_m)} \right]^2$$

- *Equal Diffusivity Limit* ($Pr = 1, \beta_m = \beta_h = \beta$): Reduces seamlessly to the smooth quadratic formulation for $0 \leq Ri \leq Ri_c$:

$$S_m(Ri) = S_h(Ri) = \left(1 - \frac{Ri}{Ri_c}\right)^2$$

- **Unstable Boundary Layer ($\zeta < 0$) Solutions:**
- *Master Cubic Polynomial*: Governed in variable $x \equiv S_m^2(Ri)$ by:

$$x^3 - 2x^2 + (1 - Ri^2\beta_m\beta_h)x - Ri^2\beta_m(\beta_m - \beta_h) = 0$$

- *Equal Coefficient Case* ($\beta_m = \beta_h = 16$): Collapses back to the widely implemented operational form:

$$S_m(Ri) = (1 - 16Ri)^{1/2}$$

Merge & Errata Verification

- **Transformation Integrity:** Equation (17) is the essential fix; the exponent $-3/2$ on $S_m(Ri)$ corrects the dimensional consistency when $Pr \neq 1$.
- **Symbol Substitution:** Ensure all instances of legacy $f_m(Ri)$ and $f_h(Ri)$ in text blocks are globally replaced, especially in context near critical Richardson number Ri_c derivations.